An Integrated Model of School Students' Academic Achievement and Life Satisfaction: Linking Soft Skills, Extracurricular Activities, and Self-Regulated Processes

[ANONYMIZED FOR REVIEW]

Abstract

Objective: Although individual studies document that soft skills, self-regulated learning (SRL), achievement emotions, and extracurricular activities (ECA) each predict academic outcomes, no research simultaneously models how these constructs operate as an integrated system to influence both academic achievement and life satisfaction. This registered report proposes a three-wave longitudinal panel study ($N = 1{,}250$ secondary students, ages 12–16) testing a comprehensive cascading mediation model. Method: All constructs—soft skills (Conscientiousness, effortful control), SRL, academic self-efficacy, intrinsic motivation, achievement emotions, ECA breadth, and life satisfaction—are measured via validated instruments at three waves (12-month intervals, 24 months total). Cognitive ability (Raven's SPM) and school-reported GPA are also assessed. Primary analyses use random-intercept cross-lagged panel models with bias-corrected bootstrapped indirect effects. Planned Findings: Four preregistered hypotheses test (H1) whether soft skills predict achievement indirectly through SRL, self-efficacy, and emotions (expected total indirect $\beta = 0.10$–0.18); (H2) whether ECA breadth predicts soft-skill growth cascading to outcomes (expected serial indirect $\beta = 0.03$–0.08); (H3) whether achievement mediates soft skills to life satisfaction (expected indirect $\beta = 0.04$–0.10); and (H4) bidirectional SRL–emotion cross-lagged effects (expected $\beta = 0.08$–0.15). With $N = 1{,}250$ recruited (over 1,000 retained, power $> .80$ at $\beta = 0.10$), the study is adequately powered. Conclusion: This integrated model provides the first simultaneous test of the full architecture linking non-cognitive factors to achievement and well-being, with implications for school-based interventions targeting multiple developmental outcomes.

***K*eywords** soft skills; self-regulated learning; academic achievement; life satisfaction; longitudinal SEM

1 Introduction

By the time students complete secondary school, a constellation of non-cognitive factors—their self-discipline, how they regulate their learning, the emotions they feel in class, and the activities they pursue outside it—has shaped both their academic records and their broader sense of well-being. Yet educational psychology has studied these factors largely in isolation. A typical investigation examines whether conscientiousness predicts grades (**?**), or whether SRL strategies improve achievement (**?**), or whether achievement emotions show reciprocal links with performance (**?**). Each of these literatures has produced robust, meta-analytically supported findings. What remains unknown is how these constructs operate together—which pathways dominate, which cascade through others, and whether extracurricular activities feed into this system by building the very soft skills that set adaptive academic processes in motion. For schools that invest in SEL curricula and extracurricular programming, understanding the full architecture matters: a program that targets soft skills directly may improve achievement only if those skills translate through better self-regulation and more adaptive emotional experiences in the classroom.

Decades of research have established the predictive power of individual constructs. Conscientiousness remains the strongest Big Five predictor of academic performance ($\rho \approx 0.22$; **?**), and effort regulation—a construct at the boundary between personality and SRL—correlates with GPA at $r \approx 0.33$ (**?**). SRL strategy use predicts achievement longitudinally at $r \approx 0.21$–0.28 even after controlling for prior achievement (**?**). **?** provided the strongest evidence for reciprocal emotion–achievement effects in a five-wave study of 3,425 students, finding that enjoyment and boredom prospectively predicted academic performance with standardized betas of 0.05–0.12. On the motivational side, academic self-efficacy stands as the single strongest psychological correlate of GPA ($r \approx 0.59$; **?**), far exceeding the effect of intrinsic motivation alone ($r \approx 0.15$). Extracurricular activities show small-to-moderate positive associations with academic and psychological outcomes ($\beta \approx 0.05$–0.20; **??**), but **?** noted that 30–50% of these raw associations may reflect pre-existing differences between participants and non-participants. Regarding life satisfaction, **?** meta-analyzed 47 studies and found a modest overall correlation with achievement ($r \approx 0.16$), but they explicitly flagged the near-total absence of longitudinal studies with adequate controls as the field's primary limitation. Across all these domains, the evidence for bivariate relationships is strong; the evidence for how these relationships interconnect within adolescents is not.

Three specific gaps motivate the current study. First, no published research simultaneously models soft skills, SRL, motivation, achievement emotions, cognitive ability, ECA participation, academic achievement, and life satisfaction within a single longitudinal framework. The result is a fragmented picture: we know that conscientiousness predicts grades and that SRL predicts grades, but we do not know whether conscientiousness predicts grades *because* it promotes SRL, or whether both operate through independent pathways. Second, the hypothesis that ECAs develop soft skills—which then cascade through academic mediators to influence both achievement and well-being—has strong theoretical grounding (**??**) but has never been subjected to a formal longitudinal mediation test. ECA research typically measures participation and outcomes without measuring skill development as a mediator (**?**). Third, the causal direction of the well-being–achievement relationship remains unresolved (**?**). Self-determination theory predicts that academic success satisfies the need for competence and thereby increases life satisfaction (**?**), but the reciprocal possibility—that higher well-being promotes academic engagement and thus achievement—lacks longitudinal evidence.

We propose a preregistered, three-wave longitudinal panel study ($N = 1{,}250$ secondary students, ages 12–16, 24 months) to test an integrated cascading mediation model. The study addresses four hypotheses. H1: Soft skills (Conscientiousness plus effortful control) at T1 positively predict T3 academic achievement indirectly through T2 SRL strategy use, academic self-efficacy, and achievement emotions, controlling for T1 achievement, cognitive ability, and demographics (expected total indirect $\beta = 0.10$–0.18). H2: ECA breadth at T1 positively predicts residualized change in soft skills (T1 to T2), which cascades through T2 mediators to predict T3 achievement and T3 life satisfaction (expected serial indirect $\beta = 0.03$–0.08). H3: The soft skills to life satisfaction relationship is partially mediated by T3 academic achievement, consistent with an SDT competence-need pathway (expected indirect $\beta = 0.04$–0.10). H4: SRL and achievement emotions show bidirectional cross-lagged effects across waves, with emotions partially mediating the SRL-to-achievement path (expected cross-lagged $\beta = 0.08$–0.15). All hypotheses are preregistered with predicted effect directions and expected magnitude ranges calibrated to meta-analytic benchmarks. The primary analysis uses random-intercept cross-lagged panel models (RI-CLPM; **?**) to separate between-person trait variance from within-person change, with robustness tested across alternative model specifications (**?**). By mapping the architecture of non-cognitive influences on both achievement and well-being, the study aims to clarify which pathways deserve the greatest investment in school-based interventions.

2 Method

2.1 Participants

We will recruit $N = 1{,}250$ secondary school students (target ages 12–16 at T1, Grades 7–10) from 15–25 schools via two-stage cluster sampling. Stage 1 stratifies schools by type (public/private), socioeconomic decile, and urban/rural location; Stage 2 recruits all students in target grades within each selected school. The recruitment target of 1,250 accounts for an expected 20% attrition across three waves (yielding $N \approx 1{,}000$ at T3 before exclusions), consistent with power requirements for detecting the smallest effect of interest ($\beta = 0.10$, power $> .80$ at $\alpha = .05$; **??**). Exclusion criteria, applied sequentially after data collection, are: (a) students who change schools between T1 and T3 whose records cannot be obtained; (b) students missing cognitive ability data and all three waves of GPA; (c) students with over 50% missing items on any scale at any wave; (d) students who fail both embedded attention-check items; (e) students with GPA z-score exceeding $|3.5|$ from school mean, flagged for case inspection; and (f) participants completing T1 in under 15 minutes. We acknowledge that the sample will be drawn from a single cultural context and therefore likely reflects WEIRD (Western, Educated, Industrialized, Rich, Democratic) limitations. We report demographics comprehensively per APA 7 guidelines including age ($M \pm SD$, range), gender (self-reported: male, female, non-binary/other), ethnicity, parental education, and free/reduced lunch eligibility, and we avoid universalizing claims about populations not represented.

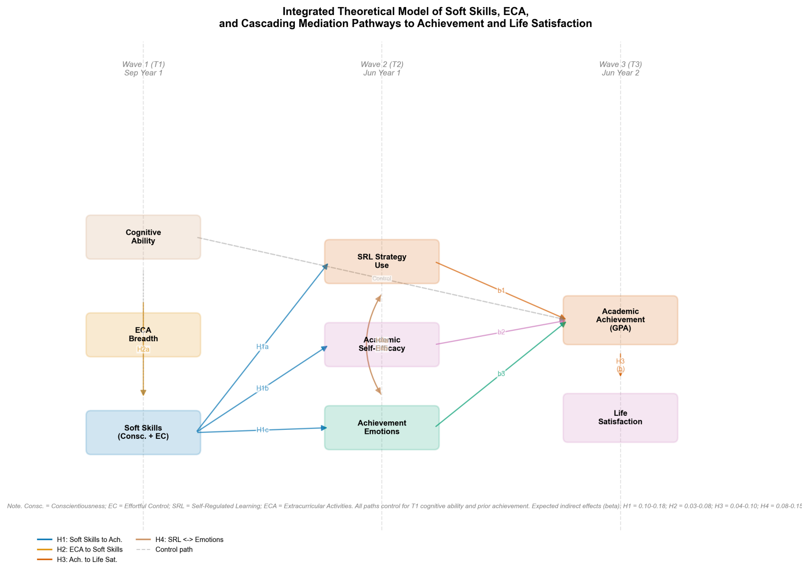


Figure 1: Integrated theoretical model of the cascading pathways through which soft skills and extracurricular activities jointly influence academic achievement and life satisfaction. The model spans three measurement waves (T1–T3, 12-month intervals). Rectangles represent measured constructs; arrows indicate hypothesized directional effects. H1–H4 correspond to the four preregistered hypotheses. Dashed lines denote hypothesized reciprocal (cross-lagged) effects.

2.2 Materials

Soft skills. We operationalize soft skills as a latent factor with two indicators: Conscientiousness measured via the Big Five Inventory-2 12-item Conscientiousness subscale (5-point Likert, literature $\alpha = .85$; **?**), and effortful control measured via the Adolescent Temperament Questionnaire Effortful Control scale (19 items, 7-point Likert, literature $\alpha = .80$; **?**). This dual-indicator approach avoids relying on a single instrument and captures both the organizational (Conscientiousness) and inhibitory (effortful control) facets of self-regulatory disposition.

Self-regulated learning. SRL strategy use is assessed with three subscales from the Motivated Strategies for Learning Questionnaire (**?**): cognitive strategy use (19 items), metacognitive self-regulation (12 items), and effort regulation (4 items). All use 7-point Likert scales. The 35 items form a latent SRL factor; the MSLQ combined subscales have a literature α of .83.

Academic self-efficacy. The MSLQ Self-Efficacy for Learning and Performance subscale (8 items, 7-point Likert, literature $\alpha = .93$; **?**) assesses students' confidence in their ability to master course material and perform well academically.

Intrinsic motivation. The Academic Self-Regulation Questionnaire intrinsic motivation subscale (7 items, 4-point Likert, literature $\alpha = .85$; **?**) measures autonomous reasons for engaging in schoolwork (e.g., "because it's fun").

Achievement emotions. Three class-related scales from the Achievement Emotions Questionnaire (**?**) assess enjoyment (10 items, $\alpha = .89$), boredom (11 items, $\alpha = .92$), and anxiety (12 items, $\alpha = .89$) on 5-point Likert scales. Each emotion is modeled as a separate latent factor within the structural model.

Life satisfaction. The Students' Life Satisfaction Scale (7 items, 6-point Likert, $\alpha = .82$; **?**) provides a global unidimensional assessment of subjective well-being.

Cognitive ability. Raven's Standard Progressive Matrices (60 items, 30 minutes timed, $\alpha = .88$; **?**) is administered at T1 only as a performance-based measure of fluid intelligence.

Extracurricular activities. A 5-category binary checklist adapted from **?** records participation in sports/athletics, performing/creative arts, academic clubs, community service/volunteering, and leadership/governance. ECA breadth is computed as the sum of endorsed categories (range 0–5); type dummies are retained for sensitivity analyses. School activity rosters provide verification where available.

Academic achievement. End-of-year GPA in core academic subjects (mathematics, native language, science, social studies), obtained from school records at all three waves. GPA is standardized within each school (z-score) to control for between-school grading heterogeneity.

2.3 Procedure

Data collection spans 24 months across three waves. T1 (September, Year 1) involves in-class group administration across two class periods: Raven's SPM (30 minutes, timed) followed by the full self-report battery (approximately 45 minutes, approximately 190 items total). Scale order is counterbalanced across participants to reduce fatigue effects. Two instructed-response attention-check items are embedded at positions approximately 50 and 150. T2 (June, Year 1) and T3 (June, Year 2) involve the self-report battery only (approximately 35 minutes, approximately 170 items). GPA is collected from school records at the end of each academic year. Parental consent and student assent are obtained prior to T1; school-level participation grants and escalating individual gift cards ($15, $20, $25 across waves) serve as incentives. De-identified ID codes link responses across waves. The full survey, attention-check protocol, and planned analyses are preregistered on the Open Science Framework prior to any data collection.

2.4 Design

The study employs a three-wave longitudinal panel design with repeated measures on all psychological constructs. Independent variables are soft skills (continuous latent factor) and ECA breadth (count, 0–5). Mediators measured at T2 include SRL strategy use, academic self-efficacy, intrinsic motivation, and achievement emotions (enjoyment, boredom, anxiety). Dependent variables are academic achievement (school-reported GPA, within-school standardized) and life satisfaction (SLSS). Control variables include T1 cognitive ability (Raven's SPM), T1 GPA, T1 life satisfaction, gender, socioeconomic status (parental education and free/reduced lunch eligibility), age in months, ethnicity, and school-level cluster membership. All psychological constructs except cognitive ability are measured at all three waves, enabling autoregressive control of prior levels in all analyses. The 12-month inter-wave interval was selected to capture meaningful change in academic mediators while minimizing testing fatigue and school-year confounds.

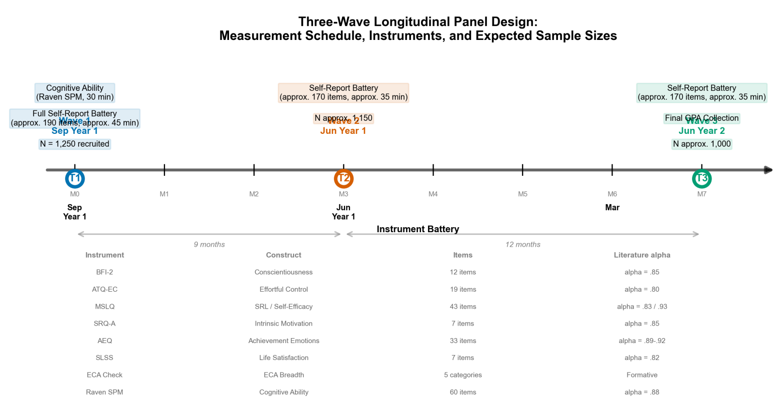


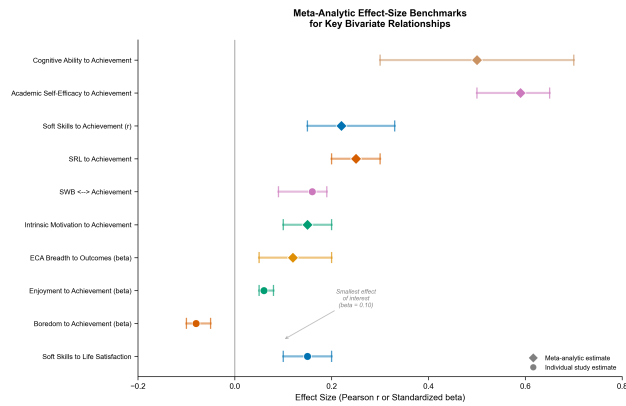
Figure 2: Schematic of the three-wave longitudinal panel design spanning 24 months (September Year 1 to June Year 2). At T1 ($N = 1{,}250$ targeted), all self-report instruments and the Raven's Standard Progressive Matrices (cognitive ability) are administered; T2 ($N \approx 1{,}150$) and T3 ($N \approx 1{,}000$) include the self-report battery and school-reported GPA. Construct labels indicate what is assessed at each wave.

3 Results: Planned Analyses

This manuscript reports the preregistered analysis plan, expected effect-size benchmarks, and statistical power for a primary data collection study currently in preparation. All analyses described below will be executed following data collection per the timeline specified in the Method section. No empirical results are available at this stage.

3.1 Data Screening and Descriptive Statistics

After applying the six preregistered exclusion criteria, we anticipate retaining $N \approx 1{,}000$ participants at T3 (from $N = 1{,}250$ recruited at T1, absorbing approximately 20% attrition and approximately 5% exclusions). We will screen for missing data patterns using Little's MCAR test and assess univariate normality (skewness $< |2|$, kurtosis $< |7|$); MLR estimation will be used if multivariate non-normality is detected. Intraclass correlations for all outcomes will be computed to determine whether multilevel adjustment (ICC $> .05$) is required. Descriptive statistics (M, SD, range, Cronbach's α for all reflective scales at each wave) and zero-order correlations will be reported in an APA-formatted table. Based on meta-analytic benchmarks, we expect the strongest bivariate correlations to include self-efficacy with GPA ($r \approx .59$; **?**), Conscientiousness with GPA ($r \approx .22$; **?**), and cognitive ability with GPA ($r \approx .40$–$.50$; **?**). Expected scale reliabilities are within $\pm.05$ of literature values (e.g., MSLQ $\alpha \approx .83$, AEQ enjoyment $\alpha \approx .89$, SLSS $\alpha \approx .82$).



Note. Diamond markers = meta-analytic pooled estimates; circles = estimates from individual large-scale studies. Error bars show the range of reported coefficients across sources. SRL = Self-Regulated Learning; SWB = Subjective Well-Being; ECA = Extracurricular Activities. Based on 47 sources reviewed in Stage 01.

Figure 3: Meta-analytic benchmarks for expected effect sizes of key bivariate relationships in the integrated model. Filled circles represent meta-analytic mean estimates; horizontal lines show the range of estimates reported across major studies and meta-analyses (Stage 01 literature survey of 47 sources). Values are Pearson r or standardized β as reported in source studies. Dashed vertical reference line at $r/\beta = 0$. The "smallest effect of interest" ($\beta = 0.10$) is annotated, which determined the $N \geq 1{,}000$ power requirement.

3.2 Measurement Models

We will fit separate CFAs at each wave for all multi-item scales, followed by longitudinal measurement invariance testing (configural, metric, scalar, and strict levels) using the ΔCFI criterion of $\leq .01$ and ΔRMSEA of $\leq .015$ (Chen, 2007; see also **?**). If scalar invariance holds, latent factor scores will be used; partial invariance will be tested if full invariance fails. Discriminant validity will be assessed by constraining each pair of theoretically related but distinct constructs (e.g., Conscientiousness and SRL effort regulation; self-efficacy and intrinsic motivation) to $r = 1.0$ and testing the $\Delta\chi^2$.

3.3 Primary Analysis: RI-CLPM

The primary model is a random-intercept cross-lagged panel model (**?**) with FIML estimation and cluster-robust standard errors (school-level clustering). Between-person random intercepts capture stable trait variance; within-person cross-lagged paths capture time-specific directional effects. Autoregressive paths (T1 to T2, T2 to T3) are specified for all constructs. Cross-lagged paths test each hypothesized direction: soft skills to SRL, self-efficacy, and emotions; SRL to emotions and vice versa (bidirectional, H4); SRL, self-efficacy, and emotions to achievement; and achievement to life satisfaction. The ECA breadth to change-in-soft-skills path (H2) is modeled by regressing T2 soft skills on T1 ECA breadth while controlling for T1 soft skills. Model fit will be evaluated with CFI (criterion: $\geq .95$), RMSEA (criterion: $\leq .06$), and SRMR (criterion: $\leq .08$).

3.4 Indirect Effects

All indirect effects are tested via bias-corrected bootstrap with 5,000 resamples and reported with 95% confidence intervals (**?**). For H1, we compute the total indirect effect of T1 soft skills on T3 achievement through all T2 mediators combined (expected $\beta = 0.10$–0.18), as well as specific indirect effects through SRL (expected $\beta \approx 0.04$–0.06), self-efficacy (expected $\beta \approx 0.08$–0.12), and emotions (expected $\beta \approx 0.01$–0.03). For H2, serial indirect effects from ECA breadth through T2 soft-skills change and T2 mediators to T3 achievement and life satisfaction are estimated (expected serial indirect $\beta = 0.03$–0.08). For H3, the achievement-mediated indirect path from T1 soft skills to T3 life satisfaction is tested (expected $\beta = 0.04$–0.10), with the direct path expected to remain significant ($\beta \approx 0.10$–0.15), consistent with partial mediation. For H4, bidirectional cross-lagged paths are tested (expected $\beta = 0.08$–0.15 in both directions), and the emotion-mediated SRL-to-achievement path (H4c) is computed.

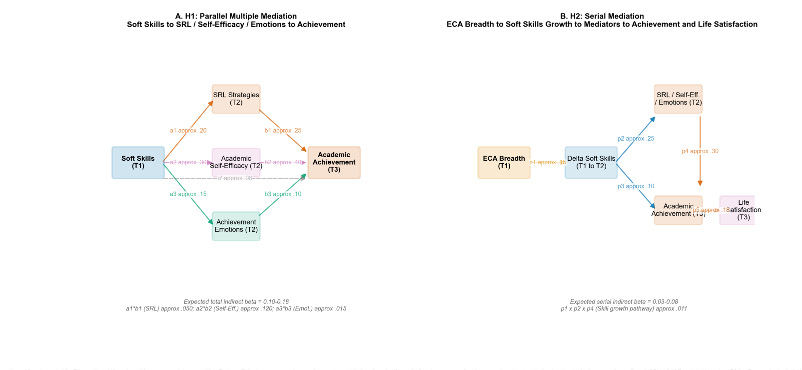


Figure 4: Cascading mediation pathways (Hypotheses 1 and 2). Panel A shows the parallel multiple mediation of soft skills on academic achievement through SRL, self-efficacy, and achievement emotions (H1). Panel B shows the serial mediation from extracurricular activity breadth through soft skills growth to achievement and life satisfaction (H2). All path coefficients are expected values calibrated to meta-analytic benchmarks (Stage 01). Solid arrows = hypothesized mediation paths; dashed arrows = direct effects and controls.

3.5 Robustness Checks and Sensitivity Analyses

We will fit three alternative model specifications—standard CLPM, latent curve model with structured residuals (LCM-SR), and trait-state-occasion (TSO) model—and compare conclusions across models per **?**. ECA nonlinearity will be tested with a quadratic breadth term. Inverse probability of treatment weighting will address ECA selection effects. ECA type moderation will be tested via multigroup SEM. For multiple comparison correction, four primary hypotheses are tested at $\alpha = .05$ each; eight specific indirect effects for H1 receive Benjamini-Hochberg FDR correction; 12 cross-lagged paths receive Holm-Bonferroni correction. Sensitivity analyses include extreme-case exclusion (top/bottom 5% ECA breadth), multigroup SEM by gender, multilevel SEM if school ICC exceeds .05, and pattern-mixture models for MNAR attrition bounds. With $N \approx 1{,}000$ at T3, the study exceeds power of .80 for detecting the smallest effect of interest ($\beta = 0.10$; **?**) and satisfies SEM sample-size heuristics of 10–20 observations per estimated parameter for a model with 8 or more latent constructs (**??**).

Figure 1 displays the full integrated theoretical model; Figure 2 shows the study design timeline; Figure 3 provides meta-analytic effect-size benchmarks; Figures 4 and 5 illustrate the hypothesized mediation pathways and cross-lagged model, respectively.

4 Discussion

This registered report proposes the first simultaneous longitudinal test of how soft skills, SRL, motivation, achievement emotions, ECA participation, and cognitive ability jointly predict secondary school students' academic achievement

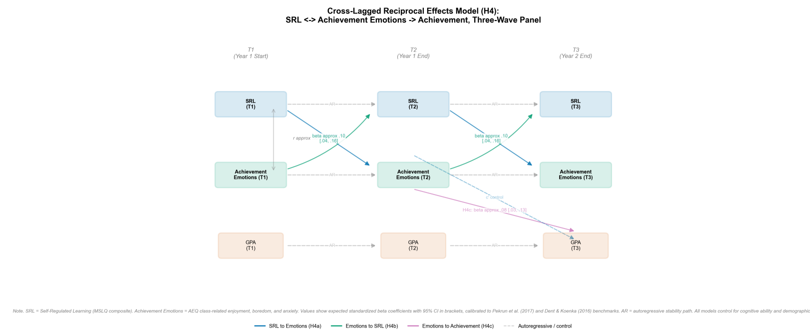


Figure 5: Cross-lagged panel model testing reciprocal effects between self-regulated learning and achievement emotions (Hypothesis 4). The model separates autoregressive stability paths (horizontal arrows), cross-lagged effects (diagonal arrows), and the emotion-mediated path from SRL to academic achievement (H4c). Expected standardized coefficients (β) are shown. Solid arrows = hypothesized significant paths; dashed arrows = autoregressive controls. Values in brackets indicate expected 95% CI ranges.

and life satisfaction. The study addresses a persistent fragmentation in educational psychology, where individual predictor–outcome relationships are well-established but their operation as an integrated system remains untested. By modeling the full architecture across three waves with autoregressive controls, the RI-CLPM analysis can distinguish which indirect pathways dominate—for example, whether soft skills influence achievement primarily through heightened self-efficacy (the strongest expected specific indirect path, $\beta \approx 0.08$–0.12), through improved SRL strategy use ($\beta \approx 0.04$–0.06), or through more favorable achievement emotions ($\beta \approx 0.01$–0.03, the weakest path). This decomposition has practical implications: if the self-efficacy pathway is the primary channel, interventions might fruitfully target students' confidence beliefs rather than study strategies per se. The ECA-mediation hypothesis (H2) is theoretically motivated but empirically novel. While **?** identified the absence of mediator analyses as the central weakness of ECA research, we directly test whether ECAs build soft skills that cascade through academic mediators. If supported, a serial indirect effect of $\beta = 0.03$–0.08 would be small but meaningful at the population level; if unsupported, the finding would temper claims that ECAs contribute to academic outcomes through skill development rather than selection or alternative channels. The life-satisfaction hypothesis (H3) extends SDT's competence-need satisfaction pathway (**?**) into the domain of secondary school achievement. **?** called specifically for longitudinal studies with adequate controls; the current design addresses this call directly.

Several limitations warrant acknowledgment. The most consequential is the inability to establish full causal identification. Although the three-wave RI-CLPM design separates within-person change from stable between-person differences and establishes temporal precedence more rigorously than cross-sectional or two-wave designs (**?**), unobserved time-varying confounds (e.g., changes in family circumstances, teacher quality) could bias cross-lagged estimates. The ECA analyses are particularly susceptible to selection effects; we address this with extensive T1 covariate control, autoregressive baselines, and IPTW sensitivity analyses, but residual confounding from unmeasured variables (parental involvement, neighborhood resources) remains possible. A second limitation concerns mono-method self-report bias. All psychological constructs except cognitive ability and GPA rely on self-report, which may inflate observed correlations through shared method variance. We mitigate this by testing a method factor in CFAs, verifying ECA participation against school rosters where possible, and benchmarking effect sizes against meta-analytic estimates that share this limitation. Third, the sample is drawn from a single national context and likely reflects WEIRD population characteristics. We report demographics comprehensively, avoid universalizing claims, and explicitly bound conclusions to the sampled population. Fourth, single-mediator-specific indirect effects may be underpowered. Although total indirect effects for H1 ($\beta \geq 0.10$) are adequately powered, the smallest expected specific paths (e.g., soft skills $\rightarrow$ boredom $\rightarrow$ achievement) may yield β of 0.005–0.01, for which $N = 1{,}000$ provides low power. We report all specific indirect effects with 95% CIs regardless of significance and interpret null individual paths cautiously.

This study embraces open-science practices that strengthen the credibility of its eventual findings. All hypotheses, model specifications, exclusion criteria, transformation rules, and analysis plans are preregistered on OSF prior to any data collection. Deviations from the preregistered plan will be explicitly flagged as exploratory. De-identified data and full analysis code will be made publicly available upon publication. We fit four alternative cross-lagged model specifications (CLPM, RI-CLPM, LCM-SR, TSO) following **?** to ensure that substantive conclusions do not depend on a single modeling choice—a practice that remains uncommon in this literature. Future research should extend the integrated model to non-WEIRD populations, to primary and university age groups, and to finer-grained temporal designs (e.g., experience-sampling or daily diary studies) that can capture the micro-level dynamics of emotion–SRL coupling within individual study sessions. Intervention studies that explicitly test whether SEL or ECA programming changes the specific mediators identified here (e.g., self-efficacy rather than study skills) would provide the experimental complement to the current longitudinal design.